COURSE 3 · PHYSICS

Physics

Printable flashcards

178 cards: equations, key ideas, quick-check questions and problems.
Cut, fold and review; or use the scheduled version in the app.

How to use

Printable flashcards

Each row is one card: the prompt on the left, the answer on the right. Print single-sided, cut along the rows, and fold each strip down the middle so the answer sits on the back. The same cards are in the app, where they are scheduled for review automatically.

Module 3.1 · Motion (kinematics)

EQUATION Write the equation: Constant-acceleration equations	$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{(u+v)t}{2}$
EQUATION Write the equation: Launch components	$u_x = u\cos\theta, \qquad u_y = u\sin\theta$
EQUATION Write the equation: Range on level ground	$R = \frac{u^2 \sin 2\theta}{g}$
KEY IDEA Complete the key idea: Gradient of x–t is velocity; gradient …	Gradient of x–t is velocity; gradient of v–t is acceleration; area under v–t is displacement
KEY IDEA Complete the key idea: SUVAT equations …	SUVAT equations need constant acceleration
KEY IDEA Complete the key idea: Projectiles: horizontal and …	Projectiles: horizontal and vertical motions are independent
QUICK CHECK The area under a velocity–time graph gives	displacement
QUICK CHECK A car at constant speed around a bend is	accelerating
QUICK CHECK Ignoring air, a projectile's horizontal velocity	stays constant
QUICK CHECK Range on level ground is greatest at	45°

PROBLEM A cyclist accelerates from 2 m/s to 8 m/s in 4 s. What is her acceleration?	1.5 m/s². (8 − 2) ÷ 4 = 1.5 m/s².
PROBLEM A stone is dropped from rest. How fast is it falling after 3 s? (g = 9.8 m/s²)	29.4 m/s. v = u + at = 0 + 9.8 × 3.
PROBLEM How far does that stone fall in 3 s?	44.1 m. ½ × 9.8 × 9 = 44.1 m.
PROBLEM A car travelling at 25 m/s brakes at 5 m/s². What is its stopping distance?	62.5 m. 0 = 625 − 10s, s = 62.5 m.
PROBLEM A ball is thrown straight up at 14 m/s. How high does it go?	10 m. s = 196 ÷ 19.6 = 10 m.
PROBLEM A ball rolls off a 1.25 m high table at 3.0 m/s. How far from the table does it land?	1.515 m. Horizontal: 3.0 × 0.505 ≈ 1.5 m.
PROBLEM At the top of a ball's flight, its acceleration is	9.8 m/s² downwards. Gravity still acts. Velocity is momentarily zero, not acceleration.

Module 3.2 · Forces and Newton's laws

EQUATION Write the equation: Newton's second law	$F_{\text{net}} = ma$
EQUATION Write the equation: Weight	$W = mg$
EQUATION Write the equation: Friction	$F_f \le \mu N$
EQUATION Write the equation: Centripetal acceleration and force	$a = \frac{v^2}{r}, \qquad F = \frac{mv^2}{r}$

KEY IDEA Complete the key idea: Net force …	Net force = mass × acceleration
KEY IDEA Complete the key idea: Action–reaction pairs …	Action–reaction pairs act on different objects
KEY IDEA Complete the key idea: Free-body diagram first, then …	Free-body diagram first, then resolve and apply F = ma
KEY IDEA Complete the key idea: Circular motion needs …	Circular motion needs an inward force mv^2/r
QUICK CHECK A body moving at constant velocity has net force	zero
QUICK CHECK Doubling mass with the same force	halves acceleration
QUICK CHECK Centripetal force points	towards the centre
QUICK CHECK Third-law force pairs act on	different objects
PROBLEM What net force gives a 1200 kg car an acceleration of 2.5 m/s^2?	3000 N. F = ma = 1200 × 2.5.
PROBLEM What is the weight of a 65 kg person on Earth (g = 9.8 m/s^2)?	637 N. W = mg = 65 × 9.8.
PROBLEM A 10 kg box is pushed with 50 N across a floor where friction is 20 N. What is its acceleration?	3 m/s^2. a = 30 ÷ 10 = 3 m/s^2.
PROBLEM A 5 kg block sits on a frictionless 30° slope. What is its acceleration down the slope?	4.9 m/s^2. a = g sin 30° = 4.9 m/s^2.

PROBLEM A crate of mass 40 kg rests on the floor. The coefficient of friction is 0.30. What horizontal force is needed to start it sliding?	117.6 N. F = μN = 0.30 × 392 ≈ 118 N.
PROBLEM A 0.5 kg ball on a 0.8 m string is swung in a horizontal circle at 4 m/s. What is the tension (ignore gravity)?	10 N. 0.5 × 16 ÷ 0.8 = 10 N.
PROBLEM A book rests on a table. The reaction partner (Newton's third law) of the book's weight is	the book's gravitational pull on Earth. The weight is Earth pulling the book. Its partner is the book pulling Earth.

Module 3.3 · Energy, work and power

EQUATION Write the equation: Work	$W = Fd\cos\theta$
EQUATION Write the equation: Kinetic energy	$E_k = \frac{1}{2}mv^2$
EQUATION Write the equation: Gravitational potential energy	$E_p = mgh$
EQUATION Write the equation: Power	$P = \frac{E}{t} = Fv$
EQUATION Write the equation: Efficiency	$\eta = \frac{E_{\text{useful}}}{E_{\text{in}}} \times 100\%$
KEY IDEA Complete the key idea: Work = force × …	Work = force × distance in the direction of the force
KEY IDEA Complete the key idea: KE = ½mv²; GPE …	KE = ½mv²; GPE = mgh; total energy is conserved
KEY IDEA Complete the key idea: Power = energy ÷ …	Power = energy ÷ time; a kWh is energy
KEY IDEA Complete the key idea: Efficiency = useful …	Efficiency = useful out ÷ total in

QUICK CHECK The unit of energy is the	joule
QUICK CHECK Doubling speed multiplies kinetic energy by	4
QUICK CHECK A kilowatt-hour measures	energy
QUICK CHECK Holding a weight still does how much work on it?	zero
PROBLEM How much work is done lifting a 20 kg box 1.5 m? (g = 9.8)	294 J. 20 × 9.8 × 1.5 = 294 J.
PROBLEM What is the kinetic energy of a 1000 kg car at 20 m/s?	2.000e+05 J. ½ × 1000 × 400 = 200 000 J.
PROBLEM A ball is dropped from 5.0 m. Ignoring air resistance, how fast is it moving when it hits the ground?	9.899 m/s. √(98) ≈ 9.9 m/s.
PROBLEM A 60 kg climber gains 300 m of height in 20 minutes. What is her average power output against gravity?	147 W. ÷ 1200 s ≈ 147 W.
PROBLEM A 2.0 kW kettle runs for 3 minutes. How many joules does it use?	3.600e+05 J. = 360 000 J.
PROBLEM A motor uses 500 J of electrical energy to lift a load, gaining 350 J of potential energy. What is its efficiency?	70 %. 350 ÷ 500 = 0.70.
PROBLEM A rope pulls a sled 10 m with 80 N at 30° to the ground. How much work is done?	692.8 J. 80 × 10 × 0.866 ≈ 693 J.

Module 3.4 · Momentum, collisions and rotation

EQUATION Write the equation: Momentum	$p = mv$
EQUATION Write the equation: Impulse	$F\Delta t = \Delta p$
EQUATION Write the equation: Conservation of momentum	$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$
EQUATION Write the equation: Torque	$\tau = Fr_\perp$
EQUATION Write the equation: Angular momentum	$L = I\omega$
KEY IDEA Complete the key idea: Momentum p = …	Momentum p = mv; impulse FΔt = Δp
KEY IDEA Complete the key idea: Total momentum is …	Total momentum is conserved without external forces
KEY IDEA Complete the key idea: Elastic collisions conserve …	Elastic collisions conserve KE; inelastic do not
KEY IDEA Complete the key idea: Torque = force × …	Torque = force × perpendicular distance; angular momentum is conserved
QUICK CHECK Momentum is	a vector
QUICK CHECK In all collisions without external forces, what is conserved?	momentum
QUICK CHECK Torque depends on force and	perpendicular distance from pivot
QUICK CHECK Impulse equals	change in momentum

PROBLEM What is the momentum of a 0.16 kg cricket ball moving at 30 m/s?	4.8 kg m/s. p = mv = 0.16 × 30.
PROBLEM A 1500 kg car at 20 m/s hits a stationary 1000 kg car and they lock together. What is their speed just after?	12 m/s. v = 30 000 ÷ 2500 = 12 m/s.
PROBLEM A 0.06 kg tennis ball arrives at 20 m/s and leaves in the opposite direction at 30 m/s. What impulse did the racket give?	3 N s. = 3.0 N s.
PROBLEM If that contact lasted 0.005 s, what was the average force?	600 N. F = Δp/Δt = 3.0 ÷ 0.005 = 600 N.
PROBLEM Airbags reduce injury mainly because they	increase collision time. Same Δp over a longer time means smaller force.
PROBLEM A 30 N force acts at right angles to a spanner 0.25 m from the bolt. What is the torque?	7.5 N m. τ = Fr = 30 × 0.25.
PROBLEM A 4.0 kg rifle fires a 0.010 kg bullet at 600 m/s. What is the rifle's recoil speed?	1.5 m/s. 4v = 0.01 × 600, v = 1.5 m/s.
PROBLEM A spinning skater pulls in her arms, halving her moment of inertia. Her rotation rate	doubles. Halve I, double ω.

Module 3.5 · Waves and sound

EQUATION Write the equation: Wave equation	$v = f\lambda, \qquad T = \frac{1}{f}$
EQUATION Write the equation: String fixed at both ends	$\lambda_n = \frac{2L}{n}, \qquad f_n = \frac{nv}{2L}$

EQUATION Write the equation: Doppler, moving source	$f' = f\frac{v}{v \mp v_s}$
KEY IDEA Complete the key idea: Transverse: perpendicular …	Transverse: perpendicular oscillation; longitudinal: parallel
KEY IDEA Complete the key idea: v = fλ; speed …	v = fλ; speed is set by the medium
KEY IDEA Complete the key idea: Waves superpose; standing …	Waves superpose; standing waves have fixed nodes
KEY IDEA Complete the key idea: Doppler: approaching …	Doppler: approaching source, higher frequency
QUICK CHECK v = fλ: doubling frequency in the same medium	halves wavelength
QUICK CHECK Destructive interference occurs when	peak meets trough
QUICK CHECK Points on a standing wave that never move are	nodes
QUICK CHECK A receding source sounds	lower
PROBLEM A wave has frequency 50 Hz and wavelength 4 m. What is its speed?	200 m/s. v = fλ = 50 × 4.
PROBLEM What is the period of a 250 Hz sound?	0.004 s. T = 1/f = 1/250.
PROBLEM Sound travels at 340 m/s. What is the wavelength of a 440 Hz note?	0.7727 m. λ = v/f = 340 ÷ 440 ≈ 0.77 m.

PROBLEM You see lightning and hear thunder 3 s later. Sound travels at 340 m/s. How far away was the strike?	1020 m. d = 340 × 3 = 1020 m.
PROBLEM A guitar string 0.65 m long carries waves at 420 m/s. What is its fundamental (n = 1) frequency?	323.1 Hz. 420 ÷ 1.3 ≈ 323 Hz.
PROBLEM Sound passes from air into water, where it travels faster. Its frequency	stays the same. Frequency is set by the source; wavelength changes instead.
PROBLEM Sound waves are	longitudinal. Air molecules oscillate along the direction of travel.

Module 3.6 · Electricity and circuits

EQUATION Write the equation: Current	$I = \frac{Q}{t}$
EQUATION Write the equation: Potential difference	$V = \frac{E}{Q}$
EQUATION Write the equation: Ohm's law	$V = IR$
EQUATION Write the equation: Series resistance	$R = R_1 + R_2 + \cdots$
EQUATION Write the equation: Parallel resistance	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$
EQUATION Write the equation: Electrical power	$P = VI = I^2R = \frac{V^2}{R}$
KEY IDEA Complete the key idea: Current = charge per …	Current = charge per second; voltage = energy per coulomb
KEY IDEA Complete the key idea: V = …	V = IR

KEY IDEA Complete the key idea: Series: R adds, current …	Series: R adds, current same. Parallel: 1/R adds, voltage same
KEY IDEA Complete the key idea: P = VI …	$P = VI = I^2R = V^2/R$
QUICK CHECK One ampere is one	coulomb per second
QUICK CHECK Adding a resistor in parallel makes the total resistance	smaller
QUICK CHECK Components in series share the same	current
QUICK CHECK A kilowatt-hour is a unit of	energy
PROBLEM A current of 2 A flows for 30 s. How much charge passes?	60 C. Q = It = 2 × 30.
PROBLEM A 12 V supply drives current through a 4 Ω resistor. What is the current?	3 A. I = V/R = 12 ÷ 4.
PROBLEM Resistors of 3 Ω, 5 Ω and 12 Ω are connected in series. What is the total resistance?	20 Ω. Add them: 3 + 5 + 12.
PROBLEM A 6 Ω and a 12 Ω resistor are connected in parallel. What is the combined resistance?	4 Ω. R = 4 Ω.
PROBLEM A 230 V heater draws 8 A. What is its power?	1840 W. P = VI = 230 × 8.
PROBLEM A 60 W lamp runs on 230 V. What is its resistance when lit?	881.7 Ω. 52 900 ÷ 60 ≈ 882 Ω.

PROBLEM A 2.4 kW appliance runs for 2.5 hours. Electricity costs 30 cents per kWh. What does it cost, in dollars?	1.8 dollars. 6 × $0.30 = $1.80.
PROBLEM Power is transmitted at higher voltage mainly to	reduce current and so I^2R heating losses. Loss I^2R falls with the square of current.

Module 3.7 · Magnetism and electromagnetism

EQUATION Write the equation: Force on a current	$F = BIL\sin\theta$
EQUATION Write the equation: Force on a moving charge	$F = qvB\sin\theta$
EQUATION Write the equation: Faraday's law	$\varepsilon = -N\frac{\Delta\Phi}{\Delta t}$
EQUATION Write the equation: Ideal transformer	$\frac{V_s}{V_p} = \frac{N_s}{N_p}$
KEY IDEA Complete the key idea: Currents create …	Currents create magnetic fields
KEY IDEA Complete the key idea: Force on a current …	Force on a current in a field: F = BIL (perpendicular)
KEY IDEA Complete the key idea: Changing magnetic …	Changing magnetic flux induces a voltage
KEY IDEA Complete the key idea: Transformers: Vs/Vp = Ns/Np; …	Transformers: Vs/Vp = Ns/Np; light is an electromagnetic wave
QUICK CHECK A current-carrying wire produces	a magnetic field
QUICK CHECK A step-down transformer has	fewer secondary turns

QUICK CHECK Induced voltage depends on	the rate of change of flux
QUICK CHECK Maxwell showed light is	an electromagnetic wave
PROBLEM A 0.5 m wire carrying 4 A lies at right angles to a 0.2 T field. What force acts on it?	0.4 N. F = BIL = 0.2 × 4 × 0.5.
PROBLEM A transformer has 500 primary turns and 25 secondary turns. With 240 V in, what is the output voltage?	12 V. Vs = Vp × Ns/Np = 240 × 25/500.
PROBLEM An electron ($q = 1.6 \times 10^{-19}$ C) moves at 2×10^{6} m/s perpendicular to a 0.01 T field. What force acts on it?	3.200e-15 N. $1.6 \times 10^{-19} \times 2 \times 10^{6} \times 0.01 = 3.2 \times 10^{-15}$ N.
PROBLEM A magnet is held still inside a coil. The induced current is	zero. Induction needs a changing flux.
PROBLEM Who first showed that an electric current produces a magnetic field?	Ørsted. Ørsted, 1820.
PROBLEM A coil of 200 turns sees its flux change by 0.03 Wb in 0.1 s. What size of voltage is induced?	60 V. $\varepsilon = N\Delta\Phi/\Delta t = 200 \times 0.03 \div 0.1 = 60$ V.
PROBLEM Transformers do not work with steady direct current because	there is no changing magnetic field. Induction requires change.
PROBLEM Explain in a few sentences how a bicycle dynamo lights a lamp.	The wheel spins a magnet near a coil (or a coil near a magnet). The magnetic flux through the coil keeps changing, which by Faraday's law induces a voltage. That voltage drives a current through the lamp. The faster you ride, the faster the flux changes, so the lamp glows brighter.

Module 3.8 · Thermodynamics and heat

EQUATION Write the equation: Kelvin and Celsius	$T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273$
EQUATION Write the equation: Specific heat	$Q = mc\,\Delta T$
EQUATION Write the equation: Latent heat	$Q = mL$
EQUATION Write the equation: First law	$\Delta U = Q - W$
EQUATION Write the equation: Maximum (Carnot) efficiency	$\eta_{\max} = 1 - \frac{T_c}{T_h}$
KEY IDEA Complete the key idea: Temperature in kelvin …	Temperature in kelvin = °C + 273
KEY IDEA Complete the key idea: Q = mcΔT for heating; …	Q = mcΔT for heating; Q = mL for change of state
KEY IDEA Complete the key idea: First law: ΔU …	First law: ΔU = Q − W
KEY IDEA Complete the key idea: Second law: heat flows hot to …	Second law: heat flows hot to cold; entropy never decreases; η ≤ 1 − Tc/Th
QUICK CHECK Absolute zero is	0 K
QUICK CHECK Heat flows spontaneously from	hot to cold
QUICK CHECK During boiling, temperature	stays constant
QUICK CHECK The first law of thermodynamics is about	conservation of energy

PROBLEM Convert 25 °C to kelvin.	298 K. 25 + 273.
PROBLEM How much energy raises 2.0 kg of water by 10 K? ($c = 4200$ J/kg K)	8.4e+04 J. $Q = mc\Delta T = 2 \times 4200 \times 10$.
PROBLEM How much energy melts 0.5 kg of ice at 0 °C? ($L = 3.34 \times 10^5$ J/kg)	1.670e+05 J. $Q = mL = 0.5 \times 3.34 \times 10^5 = 1.67 \times 10^5$ J.
PROBLEM A 2 kW kettle heats 1.5 kg of water from 20 °C to 100 °C. Ignoring losses, how long does it take?	252 s. $t = Q/P = 252$ s.
PROBLEM A gas gains 500 J of heat and does 200 J of work on its surroundings. What is the change in its internal energy?	300 J. $\Delta U = Q - W = 500 - 200$.
PROBLEM What is the maximum efficiency of an engine running between 600 K and 300 K?	50 %. $1 - 300/600 = 0.5$.
PROBLEM A refrigerator moves heat from cold to hot. Does this break the second law?	No, because work is supplied and more entropy is produced outside. The second law forbids this happening spontaneously, not with work input.
PROBLEM Explain why steam at 100 °C causes worse burns than water at 100 °C.	When steam condenses on skin it releases its latent heat of vaporisation, about 2.26 million joules per kilogram, before it even starts cooling as water. Water at 100 °C only releases heat as it cools. So the same mass of steam delivers far more energy to the skin.

Module 3.9 · Modern physics

EQUATION Write the equation: Lorentz factor and time dilation	$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}, \qquad t = \gamma t_0$
EQUATION Write the equation: Mass–energy equivalence	$E = mc^2$
EQUATION Write the equation: Photon energy	$E = hf = \frac{hc}{\lambda}$
EQUATION Write the equation: Spectral line	$\Delta E = E_{\text{upper}} - E_{\text{lower}} = hf$
EQUATION Write the equation: Radioactive decay	$N = N_0 \left(\frac{1}{2}\right)^{t/t_{1/2}}$
KEY IDEA Complete the key idea: Light speed is the same …	Light speed is the same for all observers; moving clocks run slow
KEY IDEA Complete the key idea: E = mc²: …	E = mc²: mass is concentrated energy
KEY IDEA Complete the key idea: Photons: E = …	Photons: E = hf; atoms emit specific frequencies
KEY IDEA Complete the key idea: Half-life is constant; fission …	Half-life is constant; fission and fusion release mass as energy
QUICK CHECK E = mc² says	mass and energy are equivalent
QUICK CHECK Photon energy is proportional to	frequency
QUICK CHECK After 3 half-lives, the fraction remaining is	1/8
QUICK CHECK Moving clocks	run slow

PROBLEM How much energy is equivalent to 1 g of mass? ($c = 3 \times 10^8$ m/s)	9.000e+13 J. $= 9 \times 10^{13}$ J.
PROBLEM What is the energy of a photon of green light, frequency 5.6×10^{14} Hz? ($h = 6.63 \times 10^{-34}$ J s)	3.713e-19 J. $E = hf = 6.63 \times 10^{-34} \times 5.6 \times 10^{14} \approx 3.7 \times 10^{-19}$ J.
PROBLEM A radioactive isotope has a half-life of 8 days. What fraction remains after 24 days? Give a decimal.	0.125. $(\frac{1}{2})^3 = 0.125$.
PROBLEM A sample of wood has 25% of its original carbon-14. How old is it? (half-life 5730 years)	1.146e+04 years. $2 \times 5730 = 11\,460$ years.
PROBLEM In the photoelectric effect, increasing the brightness of light below the threshold frequency	ejects no electrons. Each photon must individually have enough energy; more low-energy photons don't help.
PROBLEM The Sun produces energy mainly by	fusion of hydrogen into helium. Hydrogen fusion in the core.
PROBLEM Why does each element produce its own pattern of spectral lines?	Electrons in an atom can only occupy certain energy levels, and the spacing of those levels depends on the element's nuclear charge and electron arrangement. When an electron drops between levels it emits a photon with energy exactly equal to the gap, $E = hf$, so only certain frequencies appear. Different elements have different level spacings and so different line patterns.